

- (1 point) Select all that apply. Which of the following are hyperplanes?
A. A plane in $\mathbb{A}^3$ B. A point in $\mathbb{A}^1$ C. A line in $\mathbb{A}^3$ D. A plane in $\mathbb{A}^4$
- (1 point) For a line ℓ containing the points $M(2, 4)$ and $N(2, -5)$, parametric equations are $\begin{cases} x = \underline{\hspace{1cm}} \\ y = \underline{\hspace{1cm}} \end{cases}$. Moreover, a Cartesian equation for ℓ is $\underline{\hspace{2cm}}$.
- (1 point) For a plane π containing the point $A(1, 5, 7)$ and the coordinate axis Ox , parametric equations are $\begin{cases} x = \underline{\hspace{1cm}} \\ y = \underline{\hspace{1cm}} \\ z = \underline{\hspace{1cm}} \end{cases}$. Moreover, a Cartesian equation for π is $\underline{\hspace{2cm}}$.
- (1 point) The values a and d for which the line $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-3}{-2}$ is contained in the plane $ax + y - 2z + d = 0$ are $a = \underline{\hspace{1cm}}$, $d = \underline{\hspace{1cm}}$.
- (1 point) Select all that apply. What is the relative position of the lines $x = -3t, y = 2 + 3t, z = 1, t \in \mathbb{R}$ and $x = 1 + 5s, y = 1 + 13s, z = 1 + 10s, s \in \mathbb{R}$?
A. skew B. parallel C. coincide D. incident in one point
- (1 point) Select all that apply. The points $A(1, 0, -1)$, $B(0, 2, 3)$, $C(-2, 1, 1)$, and $D(4, 2, 3)$ are
A. coplanar B. not coplanar C. collinear D. not collinear

Setup 1: In $\mathbb{E}^3$, consider the tetrahedron $ABCD$. Let $\mathcal{K}$ be the frame with origin B and basis $(\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD})$.

- (1 point) Consider Setup 1 above. An equation of the plane containing the triangle BCD with respect to the frame $\mathcal{K}$ is $\underline{\hspace{2cm}}$.
- (1 point) Consider Setup 1 above. Cartesian equations of the line AD with respect to the frame $\mathcal{K}$ are $\begin{cases} \underline{\hspace{1cm}} \\ \underline{\hspace{1cm}} \end{cases}$.
- (1 point) Let $\mathcal{K} = (O, \mathcal{B})$ be a frame in $\mathbb{E}^2$. Let $\mathcal{K}' = (O', \mathcal{B}')$ be a second frame with $\mathcal{B}' = (\mathbf{i}', \mathbf{j}')$ and

$$[O']_{\mathcal{K}} = \begin{bmatrix} -2 \\ -2 \end{bmatrix}, \quad [\mathbf{i}']_{\mathcal{B}} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}, \quad [\mathbf{j}']_{\mathcal{B}} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

Consider the line $\ell : 3x + y - 2 = 0$ given relative to $\mathcal{K}$. Relative to $\mathcal{K}'$, a Cartesian equation for ℓ is $\underline{\hspace{2cm}}$.

Setup 2: Let $\mathcal{K} = (O, \mathcal{B})$ be a frame in $\mathbb{E}^3$. Let $\mathcal{K}' = (O', \mathcal{B}')$ be a second frame with $\mathcal{B}' = (\mathbf{i}', \mathbf{j}', \mathbf{k}')$ and

$$[O']_{\mathcal{K}} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad [\mathbf{i}']_{\mathcal{B}} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}, \quad [\mathbf{j}']_{\mathcal{B}} = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}, \quad [\mathbf{k}']_{\mathcal{B}} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

Consider the points

$$[A]_{\mathcal{K}} = \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}, \quad [B]_{\mathcal{K}} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \quad [C]_{\mathcal{K}} = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}, \quad [D]_{\mathcal{K}} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}.$$

10. (1 point) Consider Setup 2 above. Relative to $\mathcal{K}'$, parametric equations of the line CD are $\begin{cases} x' = \underline{\hspace{1cm}} \\ y' = \underline{\hspace{1cm}} \\ z' = \underline{\hspace{1cm}} \end{cases}$.
11. (1 point) Consider Setup 2 above. Relative to $\mathcal{K}'$, a Cartesian equation for the plane containing A , B , and C is $\underline{\hspace{1cm}}$.

Answers:

1. A, B

$$2. \ell : \begin{cases} x = 2 \\ y = t \end{cases}, t \in \mathbb{R} \text{ and } \ell : x = 2$$

$$3. \pi : \begin{cases} x = t + s \\ y = 5t \\ z = 7t \end{cases}, t, s \in \mathbb{R} \text{ and } \pi : 7y - 5z = 0$$

$$4. a = -2, d = 11$$

5. D

6. A, D

$$7. x + y + z = 0$$

$$8. \ell : \begin{cases} x + 1 = 0 \\ y = 0 \end{cases}$$

$$9. 8x' - y' - 10 = 0$$

$$10. \ell : \begin{cases} x' = t \\ y' = -1 + t \\ z' = t \end{cases}, t \in \mathbb{R}$$

$$11. z' = 0$$